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PHYSICS

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Center	Online	**Date**	28th June 2026

INDEX TABLE

Q. No.	Maximum Marks	Marks Obtained
1	10	
2	10	
3	10	
4	15	
5	10	
6	10	
7	20	
8	20	
9	15	

Total Marks Obtained:

Remarks:

INSTRUCTIONS

1. Do furnish the appropriate details in the answer sheet (viz. Name, Registration Number and Test Code).
2. Question No. 1 is COMPULSORY. Of the remaining four questions (Q2 to Q5), any THREE are to be attempted. Thus a candidate has to attempt FOUR questions in all, totalling 200 marks.
3. The number of marks carried by a question/part is indicated against it.
4. Answers must be written in the medium authorized in the Admission Certificate, which must be stated clearly on the cover of this Question-Cum-Answer (QCA) Booklet in the space provided. No marks will be given for answers written in medium other that the authorized one.
5. Word limit in questions, if specified, should be adhered to.
6. Any page or portion of the page left blank in the Question-Cum-Answer Booklet must be clearly struck off.

Is student recommended for One-to-One mentoring?

Recommended	**Strongly Recommended**

EVALUATION INDICATORS

1. Contextual Competence
2. Content Competence
3. Language Competence
4. Introduction Competence
5. Structure - Presentation Competence
6. Conclusion Competence

Overall Macro Comments / feedback / suggestions on Answer Booklet:

1.

2.

3.

4.

5.

6.

All the Best

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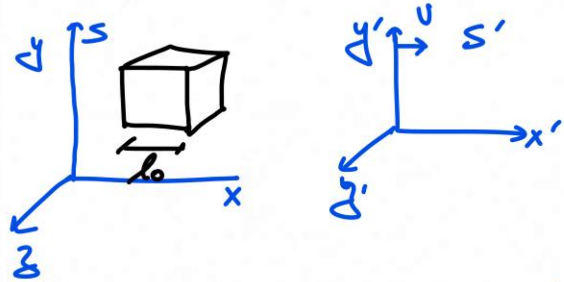
1. For a cube of proper length l_0 and proper volume V_0, show that the volume viewed from a reference frame moving with uniform velocity $\boldsymbol{u}$ in a direction parallel to an edge of the cube becomes

$$V = V_0\sqrt{1-\beta^2},$$

where $\beta = \frac{u}{c}$.

(10 Marks)

① 

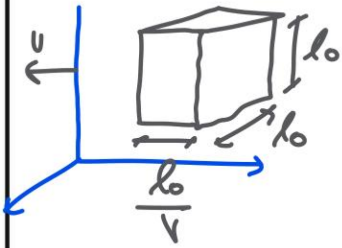
Consider a frame S' moving with 'u' speed in x direction w.r.t frame S.

l_0 = Proper length measured in S.

V_0 = Proper volume measured in S.

$\therefore V_0 = l_0^3$ — ①

Now, with respect to S', S is moving with $-u\hat{i}$. From Lorentz transformation, the length along the direction of motion contracts due to Relativity of simultaneity.



As seen from S'.

V = Volume of cube as seen from S'.

$$\therefore V = \frac{l_0^3}{\gamma} \;;\; \text{where } \gamma = \frac{1}{\sqrt{1-\frac{v^2}{c^2}}}$$

$$\therefore \boxed{V = V_0\sqrt{1-\beta^2}} \;;\; \beta = \frac{U}{c}$$

2. The spectral line of $\lambda = 5000$ Å in the light coming from a distant star is observed at 5200 Å. Find the recessional velocity of the star.

(10 Marks)

(2) The observed frequency of receding star as per doppler's Red shift is

$$\boxed{\nu = \nu_0 \sqrt{\frac{1-\beta}{1+\beta}}}$$

$\nu \downarrow$ as the star receds.

ν_0 = Frequency of emitted light in star's rest frame.

$\nu = \frac{c}{\lambda}$ $\therefore$ $\frac{1}{\lambda} = \frac{1}{\lambda_0}\sqrt{\frac{1-\beta}{1+\beta}}$

$$\therefore \boxed{\lambda = \lambda_0 \sqrt{\frac{1+\beta}{1-\beta}}}$$

Given $\lambda = 5200$Å, $\lambda_0 = 5000$Å

Since the frequency decreases $\Rightarrow$ Wavelength increases.

$$5200 = 5000\sqrt{\frac{1+\beta}{1-\beta}}$$

$$\therefore \frac{26}{25} = \sqrt{\frac{1+\beta}{1-\beta}}$$

$$\frac{1+\beta}{1-\beta} = \left(\frac{26}{25}\right)^2 = 1.08$$

$$\Rightarrow \quad 1+\beta = 1.08 - 1.08\beta$$

$$\therefore \quad 2.08\beta = 0.08$$

$$\beta = \frac{8}{208} \simeq 0.038$$

or $\boxed{\beta = 0.04}$

Since $\beta = \frac{v}{c} = 0.04 \Rightarrow \boxed{v = 0.04c}$

$\boxed{v = 12 \times 10^6 \, m/s}$

3. With what velocity should a spaceship fly so that every day spent on it may correspond to three days on Earth's surface?

(10 Marks)

3

Attaching the frame S' to spaceship.
Considering earth to be S.
∴ with respect to earth the clock of spaceship is dilated i.e.
if τ_0 = Proper time as measured in spaceship
= 1 day
∴ The time measured in earth's frame is
$t = 3$ days.
$t = \gamma\tau$ (Using Time dilation via Lorentz transformation).

$\therefore \quad \gamma\tau = 3 \text{ days}, \quad \tau = 1 \text{ day}$

$\therefore \quad \gamma = 3$

$\therefore \quad \frac{1}{\sqrt{1-\frac{v^2}{c^2}}} = 3$ or $\sqrt{1-\frac{v^2}{c^2}} = \frac{1}{3}$

$\Rightarrow 1 - \frac{v^2}{c^2} = \frac{1}{9}$

$\therefore \frac{v^2}{c^2} = 1 - \frac{1}{9} \Rightarrow \frac{v^2}{c^2} = \frac{8}{9}$

$\therefore \boxed{v = \frac{2\sqrt{2}c}{3}}$

$\therefore \boxed{v = 0.94c = 2.82 \times 10^8 \, m/s}$

4. A meson of rest mass π disintegrates into a meson of rest mass μ and a neutrino of zero rest mass. Show that the kinetic energy of the motion of the muon is

$$T = \frac{(\pi - \mu)^2 c^2}{2\pi}.$$

(15 Marks)

4

$\boxed{\text{Before decay}}$

At rest

Meson (π)

$\boxed{\text{After decay}}$

p_1 $\leftarrow \mu$ (Meson)

$\nu \rightarrow p_2$ (Neutrino)

Since there is no external force acting on the system, the linear momentum remains conserved before & after the decay.

$\therefore \boxed{p_1 = p_2}$ Both neutrino & meson carry same momentum in opposite direction.

$\therefore \; p_1 = p_2 = p \quad -①$

Conservation of Energy

$E = mc^2 = \gamma m_0 c^2$

$E_{\text{before decay}} = E_{\text{after decay}}$

$$m_\pi c^2 = \underbrace{pc}_{\text{Total energy of Neutrino (as } m_0=0)} + \underbrace{\sqrt{p^2c^2 + m_\nu^2 c^4}}_{\text{Total energy of Neutrino after decay}} \quad -(2)$$

In relativistic velocities, even a massless particle can have momentum, provided that it travels with the speed of light.

The momentum of ν & meson is taken as p (from eqn (1)).

Substituting $m_\pi = \pi$ & $m_\nu = \mu$ in eqn (2), we get,

$$\pi c^2 = pc + \sqrt{p^2c^2 + \mu^2 c^4}$$

$$\therefore \quad \pi c^2 - pc = \sqrt{p^2c^2 + \mu^2 c^4}$$

$$(\pi c^2 - pc)^2 = p^2c^2 + \mu^2 c^4$$

$$\pi^2 c^4 + \cancel{p^2c^2} - 2p\pi c^3 = \cancel{p^2c^2} + \mu^2 c^4$$

$$\therefore \quad 2p\pi c^3 = c^4(\pi^2 - \mu^2)$$

$$\therefore \quad \boxed{p = \frac{c(\pi^2 - \mu^2)}{2\pi}} \quad -(3)$$

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Now, Total energy of the muon is

$$E_{muon} = E_{initial} - \underbrace{pc}_{\text{Total energy}}$$

$$= \pi c^2 - pc$$

$$= \pi c^2 - \frac{c^2(\pi^2 - \mu^2)}{2\pi}$$

$$E_m = \frac{c^2}{2\pi}\left[2\pi^2 - \pi^2 + \mu^2\right]$$

Kinetic energy of muon $= E_{muon} - \mu c^2$

$$K = \frac{c^2}{2\pi}\left[\pi^2 + \mu^2\right] - \mu c^2$$

$$K = \frac{c^2}{2\pi}\left[\pi^2 + \mu^2 - 2\mu\pi\right]$$

$$\therefore \boxed{K = \frac{(\pi - \mu)^2 c^2}{2\pi}}$$

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5. How fast must an unstable particle move to travel 20 m before it decays, if the mean life of the particle at rest is

$$2.6 \times 10^{-8} \text{ s ?}$$

(10 Marks)

5 $\tau = 2.6 \times 10^{-8}$s is the mean life in particle's rest frame ⇒ Proper time.

For an stationary observer, the particle's clock is dilated ∴

$$\boxed{t = \gamma\tau} \text{ — (1)} \; ; \; \gamma = \frac{1}{\sqrt{1-\beta^2}} \; ; \; \beta = \frac{v}{c}$$

Also, $\boxed{vt = 20\text{m}}$ — (2)

∴ from (1), $t = \frac{2.6 \times 10^{-8}}{\sqrt{1-\beta^2}}$

Substituting 't' in eqⁿ (2) ⇒

$$v \cdot \times \frac{2.6 \times 10^{-8}}{\sqrt{1-\beta^2}} = 20$$

Multiplying the above eqⁿ with c on numerator & Denominator,

$$\frac{\beta}{\sqrt{1-\beta^2}} \times 2.6\cancel{\times 10^{-8}} \times 3\cancel{\times 10^{8}} = 20$$

$$\therefore \frac{\beta}{\sqrt{1-\beta^2}} = 2.56$$

$$\therefore \frac{\beta^2}{1-\beta^2} = 6.57 \Rightarrow \beta^2 = 6.57 - 6.57\beta^2$$

$$\therefore \beta^2 = \frac{657}{757} \Rightarrow \boxed{\beta = 0.93}$$

$$\therefore \boxed{V = 0.93c = 2.79 \times 10^8 \text{ m/s}}$$

6. A rocket leaves the Earth at a speed of 0.6c. A second rocket leaves the first at a speed of 0.9c with respect to the first. Calculate the speed of the second rocket with respect to Earth if:

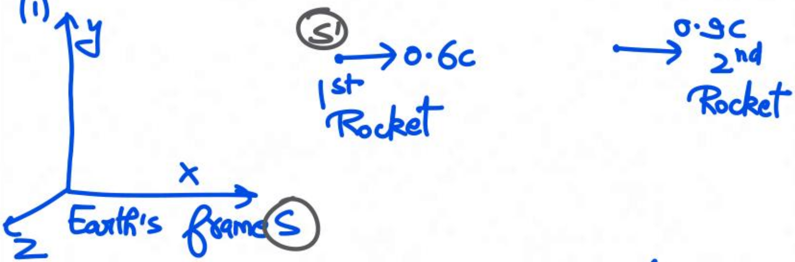
(i) It is fired in the same direction as the first one.

(ii) It is fired in a direction opposite to the first.

(10 Marks)

6

(i)



Attaching frame S' to the first rocket.

$\therefore$ $v = 0.6c$ (speed of 1st Rocket w.r.t earth).

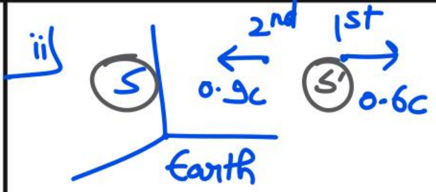
$U_x' = 0.9c$ (speed of 2nd Rocket relative to 1st).

From the Relativistic addition of velocities,

$$U_x = \frac{U_x' + v}{1 + \frac{vU_x'}{c^2}} = \frac{0.9c + 0.6c}{1 + (0.9)(0.6)} = \frac{1.5c}{1.54}$$

$\boxed{U_x = 0.97c}$ Speed of 2nd Rocket in Earth's frame.

ii)



$$\therefore U_x = \frac{U_x' + V}{1 + \frac{U_x' V}{c^2}}$$

Now, $U_x' = -0.9c$

$$\therefore U_x = \frac{-0.9c + 0.6c}{1 - 0.54} = \frac{-0.30c}{0.46} = -0.65c$$

$$\therefore \boxed{U_x = -0.65c}$$

Thus, irrespective of the source's speed, No massive object can attain the speed of light and its speed is less than c.

As the speed of rocket approaches c, its Kinetic energy increases without bounds but $v \to c$ only at ∞.

The factor $1 + \frac{U_x' V}{c^2}$ in the formula of relativistic addition maintains the Einstein's 2nd postulate of STR i.e. speed of light is same in all frames of refer. i.e 'c'.

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7. Obtain the transformations from S to S' for the components of the four-force.

(20 Marks)

8. A π^0 meson of rest mass m_0, moving with velocity u, decays in flight into two photons of the same energy. If one photon is emitted at an angle θ to the direction of the π^0 meson in the laboratory system, show that its energy is

$$h\nu = \frac{m_0 c^2}{2\gamma\left(1 - \frac{u\cos\theta}{c}\right)},$$

where

$$\gamma = \frac{1}{\sqrt{1 - \frac{u^2}{c^2}}}.$$

(20 Marks)

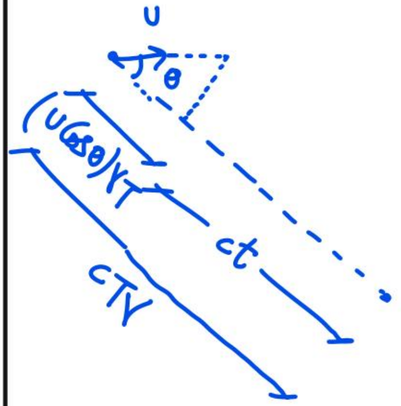
(8) Before decay | After decay

θ = Angle measured in the π-meson's rest frame.

Conservation of Relativistic energy,

$$\gamma m_0 c^2 = 2h\nu'$$

To find the frequency of emitted photon in ground frame,



$T = \frac{1}{\nu}$; where ν is freq. in lab frame.

$(v\cos\theta)\gamma T + ct = cT\gamma$

$\therefore \quad t = \frac{cT\gamma - v\cos\theta\,\gamma T}{c}$

$$\therefore \quad t = \gamma T - \frac{(v\cos\theta)}{c}\gamma T = \gamma T\left[1 - \frac{v\cos\theta}{c}\right]$$

$$\therefore \quad \boxed{\nu' = \frac{\nu}{\gamma\left(1 - \frac{v\cos\theta}{c}\right)}} \quad \text{— (1)}$$

Now, conservation of relativistic energy,

$$\gamma m_0 c^2 = 2h\nu'$$

As the energy of each photon is same

$$\therefore \quad \frac{\gamma m_0 c^2}{2h} = \frac{\nu}{\gamma\left(1 - \frac{v\cos\theta}{c^2}\right)}$$

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$$h\nu = \frac{\gamma^2 m_0 c^2 \left(1 - \frac{v\cos\theta}{c}\right)}{2}$$

9. S and S' coincide at $t = t' = 0$. S' moves with $0.6c$ along the x-axis. An event occurs at $x_1 = 10\,m,\ y_1 = 0,\ z_1 = 0,\ t_1 = 2 \times 10^{-7}\,s$, another at $x_2 = 40\,m,\ y_2 = 0,\ z_2 = 0,\ t_2 = 3 \times 10^{-7}\,s$.

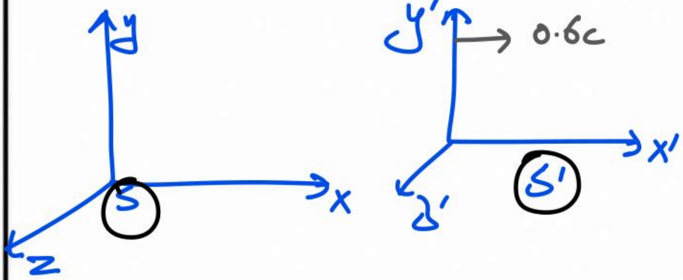
In S'

(i) What is the time difference between the events?

(ii) What is the distance between the events?

(15 Marks)

9



We have two events in S

$$\left.\begin{aligned} E_1 &: (x_1 = 10m,\ t_1 = 2\times10^{-7}s) \\ E_2 &: (x_2 = 40m,\ t_2 = 3\times10^{-7}s) \end{aligned}\right] -(1)$$

Transforming the events from S to S' using Lorentz transformation,

$$\left.\begin{aligned} x_1' &= \gamma(x_1 - vt_1) \\ x_2' &= \gamma(x_2 - vt_2) \end{aligned}\right] -(2) \quad \& \quad \left.\begin{aligned} t_1' &= \gamma\left(t_1 - \frac{vx_1}{c^2}\right) \\ t_2' &= \gamma\left(t_2 - \frac{vx_2}{c^2}\right) \end{aligned}\right] -(3)$$

(i) Time diff. b/w two events,

Using eq^n ③, we have

$$t_2'-t_1' = \gamma(t_2-t_1) - \frac{\gamma v}{c^2}(x_2-x_1)$$

Since $t_2-t_1 = 10^{-7}s$ (from ①)

$x_2-x_1 = 30m$

$\gamma = \frac{1}{\sqrt{1-\beta^2}}$; $\beta = 0.6$ $\therefore \gamma = \frac{1}{\sqrt{1-0.36}} = \frac{10}{8}$

$$\boxed{\gamma = 1.25}$$

$$\therefore \Delta t' = 1.25\times10^{-7} - \frac{1.25\times0.6}{\cancel{3}\times10^8}(\cancel{30})10$$

$$= 1.25\times10^{-7} - 0.75\times10^{-7}s$$

$$\therefore \boxed{\Delta t' = 0.5\times10^{-7}s}$$

ii) Distance between the events,

Using the invariance of spacetime interval,

$s^2 = c^2\Delta t^2 - \Delta x^2$ = Invariant across S & S'.

$\Rightarrow \quad c^2\Delta t^2 - \Delta x^2 = c^2\Delta t'^2 - \Delta x'^2$

Now, $\Delta t = 10^{-7}s$, $\Delta t' = 0.5\times10^{-7}s$

$\Delta x = 30m$.

$\therefore \Delta x'^2 = c^2[\Delta t'^2 - \Delta t^2] + \Delta x^2$

$\therefore \Delta x'^2 = 9\times10^{16}\times 1.5\times10^{-7}\times(-0.5\times10^{-7})$
$+ (30)^2$

$\Delta x'^2 = 900 - 675$

$\therefore \Delta x' = \sqrt{225}\,m$

$\therefore \boxed{\Delta x' = 15m}$